Qinyue Fei

Curriculum Vitae

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Employment

2025.07–present **Postdoctoral Fellow**, *University of Toronto*, Toronto, Canada
David A. Dunlap Department of Astronomy & Astrophysics.
Advisor: Dr. Seiji Fujimoto

Education

2019.09–2025.06 **Ph.D. in Astrophysics**, *Peking University*, Beijing, P.R. China
Department of Astronomy, Kavli Institute for Astronomy and Astrophysics (KIAA).
Advisor: Dr. Ran Wang
Thesis: *The investigation of gas dynamics in quasar host galaxies*

2023.10–2024.09 **Visiting Student**, *University of Tokyo*, Kashiwanoha, Japan
Kavli Institute for the Physics and Mathematics of the Universe (Kavli IPMU).
Advisor: Prof. John D. Silverman

2015.09–2019.06 **B.S. in Astronomy**, *Xiamen University*, Xiamen, P.R. China
Department of Astronomy. Advisor: Prof. Taotao Fang
Thesis: *Investigating the circumgalactic medium around sub-L_* galaxies in simulation*

Research Interests

- High-z massive black hole formation, supermassive black holes, and their coevolution with host galaxies
- Kinematics and dynamics of multi-phase gas in galaxies
- The properties of the interstellar medium in galaxies and their connection to star formation
- The connection between dark matter halos and galaxy evolution

Research Expertise

- Design and propose new observations for telescopes such as ALMA and *JWST*
- Data reduction and analysis of interferometric observations (ALMA, JVLA, etc.)
- Modeling gas kinematics in galaxies with 3DBarolo, Dysmalpy, GalPaK3D, qubefit, etc.
- Modeling galaxy optical images and performing SED fitting

Awarded Telescope Time

103.5 hrs in total

VLA/26A-461 **PI**, 4.0 hrs
Weighing a giant at dawn: A VLA/JWST Baryon and DM Census of a $z \sim 6$ Quasar Host

NOEMA W25EE **PI**, 19.0 hrs (co-PI: F. Walter)
Unveiling the Hidden Host of the Remarkable Metal Absorber at $z = 7.033$

ALMA 2024.1.00535.S Co-I, **leader**, 16.3 hrs (PI: J. Molina)
Firstly resolve the central molecular zone of a low-z quasar host galaxy

ALMA 2024.1.00596.S Co-I, data analysis, 6.1 hrs (PI: J. Silverman)
Resolving the influence of luminous quasars at $z = 2$ on the ISM

ALMA Co-I, **leader**, 9.5 hrs (PI: F. Bauer)
2023.1.01062.S *Probing the jet–ISM interaction in low-z radio-quiet quasars*

VLA Co-I, discussion, 38.6 hrs (PI: R. Wang)
22B-046 *A Pilot Survey of HI gas in Quasar Host Galaxies*

NOEMA Co-I, data analysis, 10 hrs (PI: R. Wang)
W20CF *Imaging the molecular gas content in low-redshift luminous AGN–starburst systems*

Service and Teaching

2020.09–2021.01 Teaching Assistant in Fundamental Astronomy, Peking University

Talks

2026.08.10 **Gravitational Lensing in Galaxy Clusters in the JWST Era**, Santander, Spain
A Systematic Census of Strongly Lensed LRDs and Their Host Galaxy Properties through VENUS Clusters

2026.08.04 **2026 EREBUS Collaboration Meeting**, Ann Arbor, Michigan, USA
Direct pathway to the Early Supermassive Black Holes: A Red Super-Eddington Quasar in a Massive Starburst Host at $z = 7.2$

2026.03.26 **Tokyo JWST Conference 2026: Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration**, Tokyo, Japan
Tracing Early Black Hole Evolution: The Low-Mass BH Population and Their Growth Pathways to Luminous Quasars

2026.03.17 **Celebrating 40 Years of Damped Lyα Systems**, Kavli IPMU, Kashiwa, Japan
Unveiling a Dense, Highly Ionized Absorber Associated with a $z = 7$ NIR-Dark Galaxy

2026.01.12 **i2i3: still linking galaxy physics from ISM to IGM scales**, Sexten, Italy — **invited talk**
Discovery of remarkably ionized, metal-dense absorber at $z > 7$

2025.11.07 **CFC meeting**, UT Austin, Texas
Path to Supermassive BHs: Evolutionary trajectory in the early Universe

2025.09.09 **TASTY talk**, University of Toronto, Toronto
A Dynamical investigation of DM of high-z quasar host galaxies

2024.07.17 **COSPAR meeting**, Busan, South Korea
The dark matter halos of two quasars through [CII] kinematics

2024.07.12 **COSMOS meeting**, oral presentation
The dark matter halos of two quasars through [CII] kinematics

2024.06.13 **Workshop on Galaxy and Black-hole evolution**, Kashiwanoha, Japan
A Dynamical investigation of Cold Gas around $z > 6$ quasars

2024.05.14 **Kavli IPMU special seminar**, Kashiwanoha, Japan
A Dynamical investigation of Cold Gas around $z > 6$ quasars

2024.03.06 **Dense Gas in Nearby Galaxies**, Osaka, Japan
Constraining the Quasar Feedback with the Hydrodynamic Equilibrium of Molecular Gas in the Low-z Quasar Hosts

Publications

First-author and contributed publications, from NASA ADS; 29 unique publications (16 refereed), 395 citations, h-index 12 as of September 2026.

First-author publications

2026 **Q. Fei**, S. Fujimoto, G. Brammer, et al., arXiv:2609.17680
A Dense, Metal-rich Absorber Driven by a Heavily Dust-obscured Galaxy: The Direct Evidence of CGM Enrichment at $z \sim 7$

2026 **Q. Fei**, S. Fujimoto, R. P. Naidu, et al., 2026, ApJ, 1003, 244
A GLIMPSE of Intermediate Mass Black Holes in the Epoch of Reionization: Witnessing the Descendants of Direct Collapse?

2026 **Q. Fei**, S. Fujimoto, G. Brammer, et al., arXiv:2602.12325
Direct Pathway to the Early Supermassive Black Holes: A Red Super-Eddington Quasar in a Massive Starburst Host at $z = 7.2$

2025 **Q. Fei**, J. Silverman, S. Fujimoto, et al., 2025, ApJ, 980, 84
Assessing the Dark Matter Content of Two Quasar Host Galaxies at $z \sim 6$ through Gas Kinematics

2024 **Q. Fei**, R. Wang, J. Molina, et al., 2024, ApJ, 976, 201
Constraining Quasar Feedback from Analysis of the Hydrostatic Equilibrium of the Molecular Gas in Their Host Galaxies

2023 **Q. Fei**, R. Wang, J. Molina, et al., 2023, ApJ, 946, 45
Dynamics of Molecular Gas in the Central Region of the Quasar I Zwicky 1

Second/Third-author publications

2026 Y. Asada, K. Inayoshi, **Q. Fei**, et al., 2026, ApJL, 1005, L22
Origins of the UV Continuum and Balmer Emission Lines in Little Red Dots: Observational Validation of Dense Gas Envelope Models Enshrouding the AGN

Contributed publications

2026 W. Q. Sun, et al. (incl. **Q. Fei**), arXiv:2609.09274
Overmassive No More: The Case for Little Red Dots Hosting Black Hole Seeds as Massive as Single Supermassive Stars

2026 J. F. V. Allingham, et al. (incl. **Q. Fei**), 2026, ApJ, 1005, 215
VENUS: Strong-lensing Model of MACS J1931.8-2635: Revealing the Farthest Multiply Imaged Supernova

2026 Y. Asada, et al. (incl. **Q. Fei**), arXiv:2607.14355
VENUS: An Ultra-faint Galaxy Hosting the Metal-poor Type II Supernova at $z = 5.13$

2026 V. Kokorev, et al. (incl. **Q. Fei**), 2026, ApJ, 1004, 153
The Deepest GLIMPSE of a Dense Gas Cocoon Enshrouding a Little Red Dot

2026 D. A. Berg, et al. (incl. **Q. Fei**), 2026, ApJ, 1003, 112
A Fleeting GLIMPSE of N/O Enrichment at Cosmic Dawn: Evidence for Wolf Rayet N Stars in a $z = 6.1$ Galaxy

2026 M. Golubchik, et al. (incl. **Q. Fei**), 2026, A&A, 710, A226
VENUS: When Red Meets Blue: A Multiply Imaged Little Red Dot with an Apparent Blue Companion behind the Galaxy Cluster Abell 383

2026 W. Q. Sun, et al. (incl. **Q. Fei**), 2026, OJAp, 9, 62505
Little Red Dot − Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot

2026 T. Y.-Y. Hsiao, et al. (incl. **Q. Fei**), arXiv:2605.06770
A Glimpse of the Low-Mass End of the Direct Mass-Metallicity Relation at $z \sim 6$–8

2026 L. J. Furtak, et al. (incl. **Q. Fei**), arXiv:2604.18696
Other Red Dots: A Possible GLIMPSE of Normal AGB Stars at Cosmic Noon through Extreme Lensing

2026 D. Korber, et al. (incl. **Q. Fei**), arXiv:2604.08687
GLIMPSED: Direct Evidence for a Fast AGN-driven Outflow from a $z = 6.64$ Little Red Dot Host Galaxy

2026 J. D. Silverman, et al. (incl. **Q. Fei**), 2026, MNRAS, 546, stag167
Under the Glare of a Luminous Quasar, the FIR Continuum Is Still an Excellent Tracer of the ISM down to the Central Kiloparsec

2025 T. S. Tanaka, et al. (incl. **Q. Fei**), 2025, ApJ, 995, 21
Discovery of a Little Red Dot Candidate at $z \gtrsim 10$ in COSMOS-Web Based on MIRI-NIRCam Selection

2025 Z. Zhang, et al. (incl. **Q. Fei**), arXiv:2512.05180
Little Red Dot Variability over a Century Reveals Black Hole Envelope via a Giant Einstein Cross

2025 S. Fujimoto, et al. (incl. **Q. Fei**), arXiv:2512.11790
GLIMPSE-D: An Exotic Balmer-Jump Object at $z = 6.20$? Revisiting Photometric Selection and the Cosmic Abundance of Pop III Galaxies

2025 S. Li, et al. (incl. **Q. Fei**), 2025, ApJL, 993, L51
Linking Electron Density with Elevated Star Formation Activity from $z = 0$ to $z = 10$

2025 H. Atek, et al. (incl. **Q. Fei**), arXiv:2511.07542
JWST's GLIMPSE: An Overview of the Deepest Probe of Early Galaxy Formation and Cosmic Reionization

2025 M. Nakane, et al. (incl. **Q. Fei**), arXiv:2511.14483
VENUS: A Strongly Lensed Clumpy Galaxy at $z \sim 11$–12 behind the Galaxy Cluster MACS J0257.1-2325

2025 Z. Liu, et al. (incl. **Q. Fei**), 2025, MNRAS, 542, 397
A PAH Deficit in the Starburst Core of a Distant Spiral Galaxy

2024 T. S. Tanaka, et al. (incl. **Q. Fei**), 2024, PASJ, 76, 1323
Crimson Behemoth: A Massive Clumpy Structure Hosting a Dusty AGN at $z = 4.91$

2024 T. S. Tanaka, et al. (incl. **Q. Fei**), arXiv:2412.14246
Hidden in Pixels. I. Discovery of Dual "Little Red Dots" Indicates Excess Clustering on Kilo-parsec Scales

2022 Y. Liu, et al. (incl. **Q. Fei**), 2022, ApJ, 929, 69
Exploring the Radio Spectral Energy Distribution of the Ultraluminous Radio-quiet Quasar SDSS J0100+2802 at Redshift 6.3

References

Prof. Luis C. Ho	lho.pku@gmail.com	*Director and University Chair Professor, KIAA, Peking University*
Dr. Ran Wang	rwangkiaa@pku.edu.cn	*Associate Professor, KIAA, Peking University*
Prof. John D. Silverman	john.silverman@ipmu.jp	*Professor, Kavli IPMU, University of Tokyo*
Dr. Seiji Fujimoto	seiji.fujimoto@utoronto.ca	*Assistant Professor, David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto*